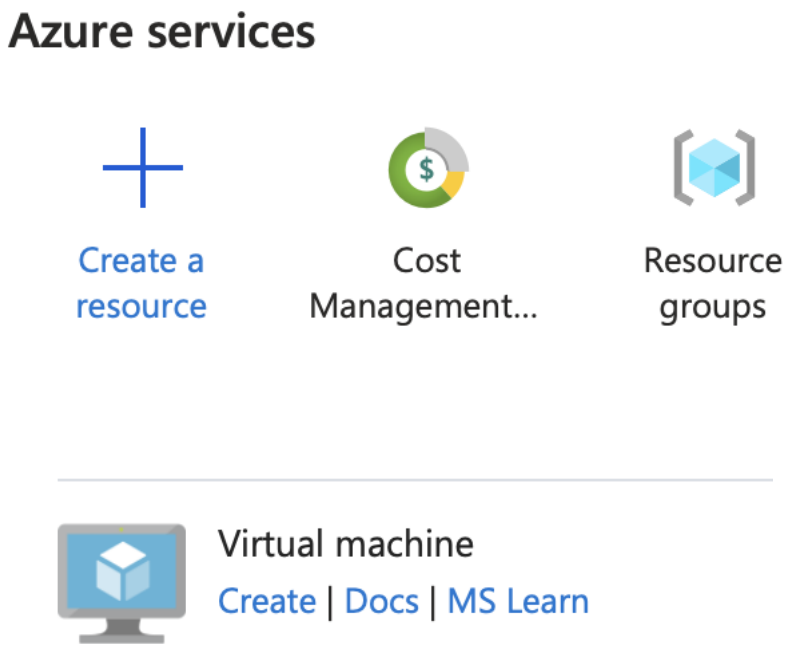
Laboratory work 01

Objective –to create IaaS resources on the Azure cloud computing platform.

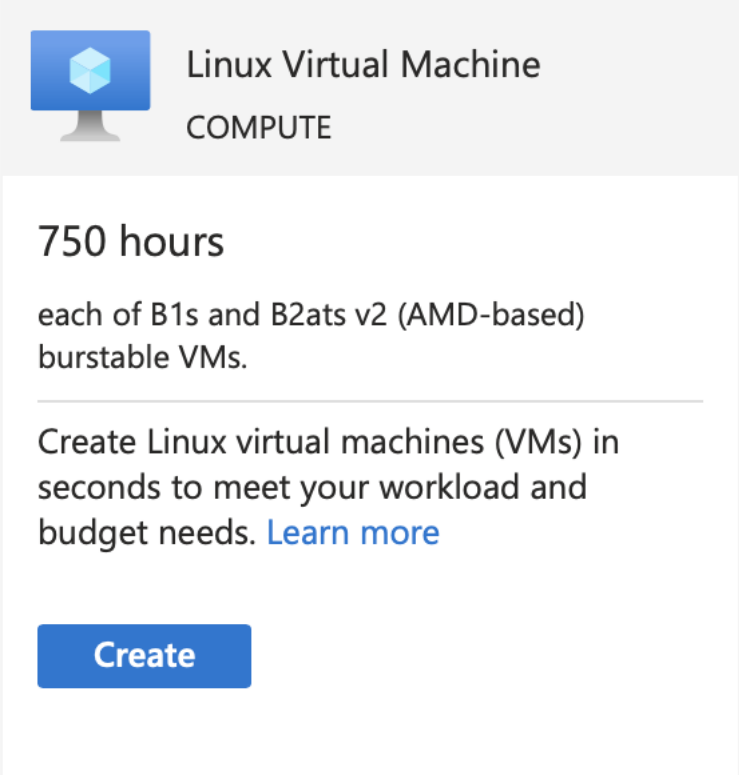
Software – portal.azure.com; Putty (or similar SSH client), draw.io, cloudflare.com

Tasks

1. Log in to the Azure portal
2. Create a virtual machine in your Azure environment, give it a name



Or go through education resourse (find free services)



3. Create a resource group - IaaS_group
4. Requirements 1 vCPU 1 GB RAM, Ubuntu 22.04 OS
5. No additional reliability requirements for infrastructure
6. You will connect to the server via SSH username/password. Create them and **remember**
7. Create a virtual network, assign a public IP address, no load balancer required
8. Select VMs to be accessed via ports 22 and 80
9. Assign an external IP address
10. Assign the TAG ENV with the value IaaS
11. Connect to the created VM using SSH and the user you created

```
sudo -i
```

12. Install the firewall and the ngninx web server:

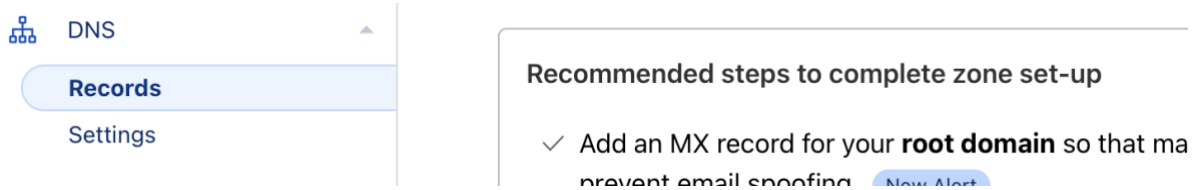
```
apt install ufw && ufw allow 22 && ufw logging off && ufw enable && ufw status
```

```
apt install nginx
```

```
ufw allow 80
```

```
systemctl status nginx
```

13. Delete default page in the directory /var/www/html
14. Using ChatGPT (or similar) create VilniusTech style front pag with your information.
15. Create a new index.html web page in the same directory
16. Test your page's reach directly via IP address
17. Disable (network security group) SSH access to the virtual machine
18. Assign a DNS name to the IP resource (resource -> settings -> configuration)
19. Test the DNS name for connectivity
20. For the domain specified in your Cloudflare account, create a DNS record (type A) leading to your-name-iaas.dclab.lt and the IP address of your VM



21. Test the DNS name for connectivity
22. Delete all resourcess

In your report, please provide:

- Work progress
- Show the resources belonging to the IaaS resource group
- A logical diagram of your solution showing VMs, IP addresses, etc.

Questions:

- What were the main steps involved in creating and configuring virtual machines in Azure?
- Did you encounter any challenges during the lab? How did you overcome them?
- How did you connect to your virtual machine? What protocols did you use, and why?
- How did you ensure the security of your virtual machines?
- Explain different ways to connect to your hosted site